order31 RESULTS

PREREG sha256 d289ca53d42172deae9568740512a68b174e984ac94a363a73484cd86699a27c (hashed before numerics). Data read-only from order30 and order28.

Controls (all PASS)

C1 stored K corner = CO: 0.0. C2 ‖R(CO)‖ = 54.406: rel 0.0. C3 arccosh(√3.5) = ½log(6+√35): 0.0.

RG25a: is the near-law generic? FAIL (Category B)

All 30 subdivision cones are non-vacuous (d′ from span{CO, Γ_γCO, vtx} between 0.34 and 0.998). 26 of 30 violate the criterion (w* = −1, |κ̂1| ≤ 0.01, r ≤ 0.10). The best weight is w* = −1 for every cone.

The four that satisfy it are exactly the cones with edge ray (1,0):

cone	™	r	κ↑
K = ((1,0),(0,1))	1.239	0.0585	0.998
((1,0),(1,1))	0.788	0.0936	0.996
((1,0),(2,1))	0.601	0.0920	0.997
((1,0),(3,1))	0.491	0.0867	0.996
((1,0),(4,1))	0.417	0.0772	0.9995

All other cones: r = 0.25–0.57 and κ̂ = 0.81 –0.97 with Im κ̂up to −0.16. The cones with edge (0,1) are the next best (r 0.25–0.30).

Prediction PASS (not blind, based on the order30 observation): **wrong**. The order30 "generic" observation came from arc-weighted sums dominated by the (1,0)-edge cones, not from single cones.

RG25b: does the miss track the arc length? Intermediate (Category B)

Pearson(r, ™) = −0.883, Spearman = −0.817. Through-origin fit r ≈ c™ has relative residual 0.92 (not proportional). Spearman(r, max|coords|) = +0.578.

The miss shrinks as the arc grows, which is the opposite of proportional. Prediction "independent" (blind): **wrong**. Verdict by rule: intermediate.

RG26: Γ_γ transport invariance: PASS (consistency with v9/v10, not new evidence)

(K2, J1): Δr = 0, Δκ̂ = 7e −18. (K1, J2): Δr = 1.4e−17, Δκ̂ = 3.3e −16.

Post-hoc reading (disclosed; not a verdict)

The near-law is attached to the edge ray e1 = (1,0) (Q(e1) = 1, the minimal vector) rather than to cones in general. Every cone with that edge has r < 0.1 and κ̂≈ 1 regardless of its arc length; cones without it do not. The negative r–™ correlation is largely explained by this, because the e1-cones are the longest at each depth.

Category, answers

Near-law generic for unimodular cone corners: **no**. Miss tracks arc length: **not proportionally**; strong negative rank correlation, mostly an e1-edge effect. Category A: none new (RG26 follows from proved transport). v11: no. Ξ: not identified. E1–E3 OPEN.

order31 NOTES

- Key new descriptive fact: near-law ⇔ cone has edge ray e1 = (1,0) (5/5 vs 0/26).
- Open next step (order32, not started): preregister RG27, "the near-law is an e1-ray effect".
  (i) Predict near-law for cones with edges e1 and (n,1), n = 5..8 (new cones).
  (ii) Predict that the γ-transported family (edge γ^{-1}e1 = (13,−2)) behaves identically (follows from RG26 logic) and that the h-image family (edge h e1 = e1) is the same family.
  (iii) Decompose the corner formula into the e1-ray B2 term, the other-ray B2 term and the B̃1B̃1 term, and test which piece carries the near-law, with thresholds fixed in advance.
  Non-vacuity span: {CO, Γ_γCO, vtx}.
- Lesson: an observation made on weighted sums does not transfer to single members; preregister member-level tests before generalising.
- Naming: no labels L1–L7 used. Geometric only; MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim; CORE_FREEZE. nice, OMP_NUM_THREADS=1, 3 GB, ~2 s; no push or sends; new files only.